

Train a Classifier

Project: Responsible Plant Classifier

1. Train a Classifier

Teach the learning cycle

Primary target: Connect training-only scaling and neighbour prediction in one pipeline

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Nearest Neighbors

Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

Look for: Compute absolute distances and rank training neighbours

Reference example, not the only acceptable wording: Distances to A,B,C,D,E are 1,1,5,7,9 cm. A and B are equally far from 3 on the numeric feature; their labels do not enter the distance calculation.

Baselines First

Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a majority label baseline using training labels only

Reference example, not the only acceptable wording: Training has four clear and two blocked, so the constant is clear. A tie chooses clear under the declared classroom rule. Validation labels score the stored prediction; using their majority to choose it would change the fitting procedure.

Sampling and Intended Use

Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

Look for: Inspect row meaning missing values and repeated records

Reference example, not the only acceptable wording: One row is intended to represent one indoor practice trip. The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Training: spots 0:H,2:H,4:S,6:S. Compare rules S if spots $\geq$ 1, $\geq$ 3, $\geq$ 5. Choose by training accuracy, then predict held-out 1:H and 5:S without revising. Report both stages.

Concept to explain

Training fits model parameters from labeled examples; prediction applies the frozen fitted model to new feature rows.

Term · Meaning

Term: fit

Meaning: learn parameters from training data

Term: parameter

Meaning: learned numeric model value

Term: predict

Meaning: apply fitted model to new features

Term: pipeline

Meaning: ordered preprocessing and model steps

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. Assemble the two previous ideas into a complete runnable program, while keeping hand tracing available without devices. Distinguish a fitted state from settings and a run record. Require students to reproduce individual predictions, including errors, rather than merely quote a score.

Readiness answer

It learns each feature minimum and range from training. Reuse the same minima, ranges, feature order and units to transform training and validation examples.

Printed lesson and workbook

Optional Python 3 editor with no packages required

Three cards labelled Fit Predict and Evaluate

Blank run record with split IDs feature order settings and outputs

Apply the connected idea: Searching candidate rules

Teach two 45-minute sessions using fictional sensor readings and reviewed labels. This is a small supervised fitting algorithm with a supplied rule form, not a complete decision-tree implementation.

Separate the prediction boundary from the candidate replacement rule. The same split and baseline principles apply, and a zero training error must never be described as proof of generalization.

Readiness answer

It chooses its constant from training label counts, using a declared tie rule. Accuracy is then measured by matching the fixed prediction to validation labels.

Printed lesson and workbook

Four training cards and four validation cards with distance and reference label

Optional Python 3 editor with no extra packages

Blank table for cutoff predictions errors and selected rule

End goal for the pictorial case

By the end, I can explain train a classifier using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Run a starter classifier and record data version, split, settings, and predictions.

MODEL IDEA Training fits model parameters from labeled examples; prediction applies the frozen fitted model to new feature rows.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Load data version v1 and a saved train/test split. Fit preprocessing on training rows only, then fit the classifier.

Draw a fresh data → model → prediction → evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 Load data version v1 and a saved train/test split.

STEP 2 Fit preprocessing on training rows only, then fit the classifier.

STEP 3 Freeze the pipeline and predict twenty test rows once.

STEP 4 Save settings, predictions, and metrics so the experiment can be reproduced.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

Training is (0,H),(1,H),(4,S),(5,S). Candidate rule predicts S for spots $\geq$ 3. Held-out cases are (2,H),(3,S).

Apply the rule and distinguish fitting from inference.

Exact explanation to model

Predictions H,S give 2/2 on this tiny set. Choosing the rule is development; applying it is inference. Two cases do not establish broad performance.

Explain the added worked example

A pipeline connects steps so a new query receives the same preparation as the training examples. This week combines the scale from Week 11 with a one-neighbour classifier. We will fit once, keep the fitted state, and call one prediction function for every raw query.

Use the same fictional trip data. Each feature pair is ordered (distance in cm, brightness in sensor units). Training IDs are A,B,C. Validation IDs V1,V2,V3 belong to separate trips; related trips were grouped before splitting. Use min-max scaling, sum of absolute scaled differences, $k=1$ and alphabetical ID order for exact distance ties.

The input contract requires two finite numeric features in the stated order and units, with no missing values. Training also needs unique IDs, known labels, at least one row and nonzero ranges. These assumptions must be checked before this short classroom program is used; its explicit guard only checks zero ranges.

fit returns minima (10,100), ranges (10,100) and scaled training examples. predict transforms a raw query with that state, ranks distances, then returns the selected ID and label. zip pairs matching features; sum adds their absolute differences. sorted uses distance first and unique ID second.

Run the whole script in Python 3. It prints ('A', 'clear'), ('C', 'blocked'), ('B', 'blocked') on separate lines. No validation labels enter fit or predict. This is a small pipeline written with functions, not the scikit-learn Pipeline class.

Key terms

Term · Meaning

Term: Pipeline

Meaning: An ordered sequence of data transformations and prediction steps

Term: Fitted state

Meaning: The learned values and stored examples needed to make predictions

Term: Reproduce

Meaning: Obtain the same result using the same recorded inputs and procedure

Term: Input contract

Meaning: The documented feature names order units and accepted values

Apply the connected idea: Searching candidate rules

Last week the baseline ignored scene features. This week a small fitting algorithm will choose a rule that uses one feature. We supply the rule form: predict blocked when a distance reading is at or below a cutoff, otherwise predict clear. The algorithm will try three cutoffs and keep the one making the fewest training mistakes.

The teacher-authored fictional rows contain a sensor distance in centimetres and a separate reviewed lane label, blocked or clear. The reading is available before prediction; the label comes from the scene review. A single distance need not describe every obstruction. These numbers are for learning, not real robot control. Each row comes from a different trip, and training and validation trips are separate.

Before scoring, declare cutoffs 3, 5 and 7 cm in ascending order, predict blocked at equality, and keep the smaller cutoff if training errors tie. This search tests only these three rules with this direction. It does not find the best possible model or prove that distance causes the label.

The output lines are 3 1, 5 0, 7 1, then chosen 5. Cutoff 3 misses the blocked row at 4 cm. Cutoff 5 matches all four training labels. Cutoff 7 wrongly marks the 6 cm clear row as blocked. Fitting stores cutoff 5; prediction then applies that stored rule to a new distance.

`best_errors` starts above any possible mistake count. The strict `<` keeps the first tied candidate; ascending order makes that the smaller cutoff. The code assumes nonempty training data, numeric distances and the two allowed labels. Validate those assumptions before using another dataset.

Guided: at 5 cm the selected rule predicts blocked because equality is included. The rule misses validation rows 2 and 4; baseline clear misses rows 1 and 4. Both score 50 percent, a zero percentage-point difference. Exit: cutoff 5 has zero mistakes on four training rows among the three supplied candidates, but scores 2/4 on validation and ties the baseline. This small comparison does not prove broad performance or global optimality.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Two stages with one fitted state

Stage · Input and action · Result

Stage: Fit once

Input and action: Training rows only

Result: Scale plus stored rows

Stage: Predict each query

Input and action: Raw pair plus state

Result: Neighbour ID and label

Stage: Evaluate afterward

Input and action: Prediction plus reference

Result: Recorded match or error

Reference labels are used after prediction to score results. They do not enter the fitted scale or stored neighbour pool.

For raw query (11,120), scale gives (0.1,0.2). Distances to A,B,C are 0.9,1.1,1.7, so the function returns A and clear. It does not recalculate minima or add the query to stored rows. The returned ID is useful for checking the reasoning, but it is not evidence that the label is true.

Trace (11,120) through both functions. What does fit store, what does predict reuse, and at which point may a validation reference label be compared?

Store the fitted state together with the input contract and method settings. A model without its scale or feature order is incomplete. Swapping the raw pair to (120,11) produces B and blocked here; the function cannot detect that the caller swapped the meanings.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Train a Classifier

Separate a reproducible fitted pipeline from merely displaying predictions.

The facts we will use

Toy training lengths are $[2, 6]$ cm with labels [healthy, spotted]. Pipeline converts cm to mm, then uses $k=1$ absolute distance. Query is 5 cm.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Train a Classifier

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fit transform

The declared conversion produces training lengths 20 and 60 mm.



2. Transform query

5 cm becomes 50 mm using the same conversion.



3. Predict neighbour

Distances 30 and 10 mm select the spotted training record.

CHANGE ONE THING

Transform training but leave query numerically at 5.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Fit transform

What happens and why: The declared conversion produces training lengths 20 and 60 mm.

Step: 2. Transform query

What happens and why: 5 cm becomes 50 mm using the same conversion.

Step: 3. Predict neighbour

What happens and why: Distances 30 and 10 mm select the spotted training record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Distances 30 and 10 mm select the spotted training record. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

A reproducible toy pipeline can still perform poorly; evaluation must use separate records and appropriate group splits.

Change one thing in the pictured case

Change: Transform training but leave query numerically at 5.

Prediction: Distances become 15 and 55, incorrectly selecting healthy.

Reason: The query is now compared in inconsistent units; the changed result is a pipeline defect.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Training is (0,H),(1,H),(4,S),(5,S). Candidate rule predicts S for spots $\geq$ 3. Held-out cases are (2,H),(3,S).

Write the question to explain: Apply the rule and distinguish fitting from inference.

Reveal after predictions: Predictions H,S give 2/2 on this tiny set. Choosing the rule is development; applying it is inference. Two cases do not establish broad performance.

Keep for an independent attempt: A classifier fits on rows 1–80 and is evaluated on those same rows. Its accuracy is 100%. What additional evidence is needed before claiming generalization?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: fit stores minima (10,100), ranges (10,100) and A (0,1) clear, B (1,0) blocked, C (1,1) blocked. predict transforms (11,120) to (0.1,0.2), measures 0.9,1.1,1.7 and returns A clear. Compare the reference only afterward. Exit: reproducible leakage is still leakage. Restore the split, fit on training only, use the frozen state for every raw validation query, then score and record all outputs.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve training-only scaling and the stored minima and ranges. State the input contract.

Stage: 7 to 18 minutes

Teacher action: Use the three cards to place data and labels at the correct stage. Complete W1.

Stage: 18 to 32 minutes

Teacher action: Run or trace the complete script. Unpack fitted state and explain zip sum and sorted.

Stage: 32 to 45 minutes

Teacher action: Use the diagram and guided question, then complete W2 from the printed outputs.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve why predict must reuse the fitted state and receive raw feature pairs.

Stage: 8 to 20 minutes

Teacher action: Complete W3 and compare two run records. Discuss why equal scores can hide different predictions.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before support. Require transformed queries and ID-linked predictions.

Stage: 35 to 45 minutes

Teacher action: Use W5 and the exit question to audit misuse. Offer the fresh retry after feedback and record support.

Teach the connected idea

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve training-majority fitting and validation scoring. Establish the distinct-trip fictional data and distance unit.

Stage: 7 to 18 minutes

Teacher action: State candidates, direction, equality and tie policy before counting. Predict all four training rows for cutoff 3 by hand.

Stage: 18 to 32 minutes

Teacher action: Trace the loop and complete W1 and W2. Name the stored parameter and distinguish fitting from later prediction.

Stage: 32 to 45 minutes

Teacher action: Reveal validation and compute both methods on the same rows. Complete the guided boundary and error-pattern question.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the difference between the prediction $\leq$ and search $<$ comparisons.

Stage: 8 to 20 minutes

Teacher action: Complete W3 and inspect both bug examples. Explain that a tie policy is a convention, not evidence of future superiority.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before hints. Require every candidate count, selected cutoff, predictions and baseline comparison.

Stage: 35 to 45 minutes

Teacher action: Use W5 and the exit claim. Give the fresh retry after feedback and record support.

Reduce help deliberately

Completion task: Give the IDs and labels printed for V1, V2, V3. Use references clear, blocked, clear to calculate accuracy and the training-majority baseline. Which error should repeat in an unchanged rerun?

Teacher check: V1 selects A clear, V2 C blocked, V3 B blocked. Accuracy is $\frac{2}{3}$. The training-majority blocked baseline is correct only for V2, scoring $\frac{1}{3}$. V3 is the repeatable error; repeated output does not turn it into a correct prediction.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for A reproducible classifier pipeline. Proposed explanation: "A pipeline prevents every leak automatically". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Correct sequencing cannot fix a fit call that receives held-out rows or features unavailable at prediction time.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: Searching candidate rules

Target: Fit a threshold rule by comparing training errors

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: List errors for cutoffs 3, 5 and 7 on the worked training rows. Identify the mistaken row for each imperfect candidate. Which cutoff is stored?

Teacher check: Errors are 1,0,1. Cutoff 3 misses row 2 at 4 cm; cutoff 7 misses row 3 at 6 cm. Cutoff 5 has no training mistakes and is stored as the best of these three candidates.

Individual check: For Searching candidate rules, use its worked case to evaluate this proposed explanation: “Zero training error means perfect predictions”. Explain the deciding step.

Expected correction: The rule was selected using those labels. New examples can disagree, as the two validation mistakes show.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

Which pipeline record would help another team reproduce the result? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

Record training inputs/labels, unit conversion, $k=1$, distance rule and query transformation; correct result is spotted from distance 10 mm. Changed-case check: Distances become 15 and 55, incorrectly selecting healthy. The query is now compared in inconsistent units; the changed result is a pipeline defect.

Changed-case reasoning: Distances become 15 and 55, incorrectly selecting healthy. The query is now compared in inconsistent units; the changed result is a pipeline defect.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Starter Classifier Run: trace code, predict outputs, execute the fixed pipeline, and compare recorded results with a peer.
3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR The model keeps learning whenever it sees test examples during evaluation.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A pipeline prevents every leak automatically: Correct sequencing cannot fix a fit call that receives held-out rows or features unavailable at prediction time.

A seed is the whole reproducibility record: This program is deterministic. Data code split feature order settings and runtime still matter; a seed cannot repair changed inputs.

Same score means same predictions: Compare predictions by example ID. Different error patterns can produce the same accuracy.

Check these misconceptions

Zero training error means perfect predictions: The rule was selected using those labels. New examples can disagree, as the two validation mistakes show.

Every comparison should include equality: Prediction and candidate replacement serve different purposes. Follow the specified boundary and tie policy separately.

The best tested rule is the best possible model: The search covers only the declared candidates and rule form, evaluated on a small set. Other forms and future cases remain unknown.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation

would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Predictions H,S give 2/2 on this tiny set. Choosing the rule is development; applying it is inference. Two cases do not establish broad performance.

Independent question

A classifier fits on rows 1–80 and is evaluated on those same rows. Its accuracy is 100%. What additional evidence is needed before claiming generalization?

Expected independent answer

Evaluate on independent examples not used to fit or tune the model, with appropriate split discipline. Training accuracy alone measures fit to seen data.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Perfect training accuracy proves perfect performance on new examples.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Threshold 1 gets 3/4, 3 gets 4/4, 5 gets 3/4. Select 3; held-out predictions are H,S, giving 2/2. This finite search is a simple training procedure. Tiny perfect test performance does not justify broad claims.

Explanations for the additional workbook examples

W1 Separate fit predict and evaluate

List what fit learns or stores, what predict takes as input, and what evaluation compares. State where the validation reference labels belong.

fit learns training minima and ranges and stores scaled training rows with IDs and labels. predict takes a raw feature pair and the fitted state. Evaluation compares a prediction with its reference label afterward. Validation references belong only in evaluation, not in fitting or the neighbour pool.

W2 Trace and score the saved run

Give the IDs and labels printed for V1,V2,V3. Use references clear,blocked,clear to calculate accuracy and the training-majority baseline. Which error should repeat in an unchanged rerun?

V1 selects A clear, V2 C blocked, V3 B blocked. Accuracy is 2/3. The training-majority blocked baseline is correct only for V2, scoring 1/3. V3 is the repeatable error; repeated output does not turn it into a correct

prediction.

W3 Repair an incomplete run record

A classmate saves only $k=1$ and a score of $2/3$. Name the information needed to reproduce this run. Explain why the same score alone is insufficient.

Record the data and code versions, exact split IDs, feature names order and units, fitted minima and ranges, stored training rows, distance and tie policy, zero-range policy, runtime version and predictions linked to validation IDs. A score hides which examples were correct; different predictions can have equal accuracy.

W4 Build and trace a fresh fitted state

Use P (0,60) blocked, Q (20,20) clear, R (20,60) clear as new training rows. Fit the same pipeline. Predict validation queries (2,28),(18,56),(4,12) with references blocked,clear,blocked. Give fitted minima and ranges, transformed queries, selected IDs, predictions, accuracy and training-majority baseline.

Minima (0,20), ranges (20,40). Scaled queries are (0.1,0.2),(0.9,0.9),(0.2,-0.2). Selected IDs P,R,Q give blocked,clear,clear. Accuracy is $2/3$; the training-majority clear baseline gets $1/3$. The third query remains wrong. Validation rows do not change the fitted state.

W5 Diagnose pipeline misuse

Explain and repair three mistakes: predict receives an already scaled query; fit receives training plus validation rows; a query uses brightness before distance. Does repeating each mistaken run make it valid?

predict expects raw features, so a pre-scaled query is scaled again; pass the raw pair. Fitting on validation leaks evaluation information; refit on training only and reevaluate. Swapped features violate the input contract; restore distance then brightness in the stated units. Repeating a mistake shows repeatability, not validity.

Searching candidate rules — W1 State the search before running it

Write the rule direction, equality behavior, candidate cutoffs and tie policy. Identify what humans supplied and what fitting selects.

Predict blocked for distance $\leq$ cutoff and clear otherwise. Try 3,5,7 cm; choose the smaller cutoff on tied training errors. Humans supplied the feature, rule form, candidates and policy; fitting selects the cutoff using training labels.

Searching candidate rules — W2 Trace the training mistakes

List errors for cutoffs 3, 5 and 7 on the worked training rows. Identify the mistaken row for each imperfect candidate. Which cutoff is stored?

Errors are 1,0,1. Cutoff 3 misses row 2 at 4 cm; cutoff 7 misses row 3 at 6 cm. Cutoff 5 has no training mistakes and is stored as the best of these three candidates.

Searching candidate rules — W3 Audit the boundary and comparison

What happens at distance 5 if prediction uses $<$ instead of $\leq$ with cutoff 5? Explain why the search uses a strict $<$ for error counts even though prediction uses $\leq$ for distance.

The changed prediction would be clear instead of blocked at 5 cm. Prediction $\leq$ includes equality in blocked. Search $<$ replaces a candidate only for fewer mistakes, preserving the first tied candidate; with ascending candidates that implements the smaller-cutoff policy.

Searching candidate rules — W4 Search and evaluate independently

New training rows are (1,blocked), (3,clear), (5,blocked), (7,clear). Try cutoffs 2,4,6, using the same direction and smaller-cutoff tie policy. Give every training error count and select a cutoff. Validation rows are (2,blocked), (4,blocked), (6,clear), (8,clear). List the selected rule's predictions and score. Compare with the training-majority baseline using clear on ties.

Training errors for 2,4,6 are 1,2,1. Choose 2 under the declared tie policy. Validation predictions are blocked,clear,clear,clear, matching rows 1,3,4: 3/4 or 75 percent. Training labels tie, so baseline clear matches rows 3,4: 2/4 or 50 percent. The selected rule gains one correct example or 25 percentage points on this small set only.

Searching candidate rules — W5 Limit the claim

The worked rule fits training perfectly but ties the baseline on validation accuracy. Explain why neither result proves it is the best possible model. Name two details to keep in the experiment record.

Perfect training fit describes the examples used for selection; the four-row validation score is only 50 percent and ties the baseline. Only three candidates of one form were searched. Keep the candidates, all error counts, tie policy, chosen cutoff, split and row-aligned validation predictions; none proves general superiority.

Assess independent transfer

Selected case: Use P (0,60) blocked, Q (20,20) clear, R (20,60) clear as new training rows. Fit the same pipeline. Predict validation queries (2,28),(18,56),(4,12) with references blocked,clear,blocked. Give fitted minima and ranges, transformed queries, selected IDs, predictions, accuracy and training-majority baseline.

Selected case answer

Minima (0,20), ranges (20,40). Scaled queries are (0.1,0.2),(0.9,0.9),(0.2,-0.2). Selected IDs P,R,Q give blocked,clear,clear. Accuracy is 2/3; the training-majority clear baseline gets 1/3. The third query remains wrong. Validation rows do not change the fitted state.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Train Plant Classifier v1 without opening the final test labels during model choice; save the experiment record.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 14: Scores and Calibration Intuition. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Label each tuple field and use one query at a time. Provide a blank fit/predict/evaluate diagram without answers. Extension: use query (10,100) in the worked model. A and B both have scaled distance 1, C has 2; alphabetical ID order selects A clear. Explain why this convention must be recorded.

Require W4 minima (0,20), ranges (20,40), scaled queries (0.1,0.2),(0.9,0.9),(0.2,-0.2), IDs P,R,Q, predictions blocked,clear,clear and scores 2/3 versus 1/3. Fresh retry: L (5,80) clear, M (15,40) blocked, N (15,80) blocked. Fit gives minima (5,40), ranges (10,40). Queries (6,48),(14,76) scale to (0.1,0.2),(0.9,0.9), selecting L clear and N blocked. Require the input contract and a statement that prediction does not refit.

Week 13 introduces confusion matrices. Keep the ID-linked reference and prediction pairs from this week. Teacher reference: https://scikit-learn.org/stable/common_pitfalls.html (consistent preprocessing and training-only fitting; reviewed 2026-09-27).

Readiness and reassessment for Searching candidate rules

Use a number line and move a cutoff marker; place equality on the blocked side. Give a blank prediction grid but no error counts. Extension: reverse tied candidate order in the supplied code and explain why the retained cutoff changes unless the tie policy is implemented explicitly.

Require W4 training errors 1,2,1, cutoff 2, validation 3/4 and baseline 2/4 with a 25 percentage-point difference. Fresh retry: training (2,blocked),(4,clear),(6,blocked),(8,clear), candidates 3,5,7. Errors are 1,2,1; select 3. Validation (3,clear),(5,blocked),(7,clear),(9,clear) gives rule blocked,clear,clear,clear: 2/4. Baseline clear gives 3/4, so the rule is one correct example or 25 percentage points worse. Require correct equality handling and a restrained claim.

Plan the delayed return

Suggested return points in this level: ai-4/14, ai-4/16, ai-4/19. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

7. Fit and test a threshold classifier

Fit and test a threshold classifier

The program chooses a threshold using training labels only. Spotted is 1 and healthy is 0. Candidate thresholds lie between consecutive training values. Test rows are used after selection.

```
"""Fit a threshold using training data, then evaluate untouched examples."""

train = [(0, 0), (1, 0), (4, 1), (5, 1)]

values = sorted({x for x, y in train})

if len(values) < 2:

    raise ValueError("Two distinct training values are required")
```

```

candidates = [(a + b) / 2 for a, b in zip(values, values[1:])]

def errors(threshold):
    return sum(int(x >= threshold) != y for x, y in train)

threshold = min(candidates, key=errors)

test = [(2, 0), (3, 1)]

predictions = [int(x >= threshold) for x, y in test]

correct = sum(pred == y for pred, (x, y) in zip(predictions, test))

print(threshold, predictions, correct, len(test))

```

Expected output: 2.5 [0, 1] 2 2

Candidate thresholds 0.5, 2.5 and 4.5 have training errors 1, 0 and 1. The selected threshold is 2.5. Both test predictions are correct. This tiny artificial example demonstrates fitting and evaluation; it does not establish reliability on real leaf photos.

Independent code change

Change the final test row to (3, 0), keeping training unchanged. Predict before running.

Teacher answer: Threshold remains 2.5 and predictions remain [0, 1], but only 1/2 test predictions is correct.

8. Prepare, teach and assess

Teaching target: Connect training-only scaling and neighbour prediction in one pipeline

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation;

40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Give the IDs and labels printed for V1, V2, V3. Use references clear, blocked, clear to calculate accuracy and the training-majority baseline. Which error should repeat in an unchanged rerun?

Reasoned practice answer: V1 selects A clear, V2 C blocked, V3 B blocked. Accuracy is 2/3. The training-majority blocked baseline is correct only for V2, scoring 1/3. V3 is the repeatable error; repeated output does not turn it into a correct prediction.

Check every learner before discussion: Use the worked case for A reproducible classifier pipeline. Proposed explanation: “A pipeline prevents every leak automatically”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Correct sequencing cannot fix a fit call that receives held-out rows or features unavailable at prediction time.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Transform training but leave query numerically at 5. Require a prediction and a reason before comparing. Expected result: Distances become 15 and 55, incorrectly selecting healthy. Reason: The query is now compared in inconsistent units; the changed result is a pipeline defect.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.